10 — Supersession Manifold

Overview

The supersession manifold detects when a new memory **replaces** (supersedes) an old one, rather than being an unrelated addition. It uses the first right singular vector (SVD1) of a multi-domain seed set of (old, new) fact pairs. The resulting direction axis is a single vector in embedding space that best explains value-substitution across domains.

Mathematical Formulation

Seed Set

A fixed set of m (old_fact, new_fact) pairs spanning 7 domains and 5 languages:

$$\mathcal{S} = \{(\text{old}_i, \text{new}_i)\}_{i=1}^{m}, \quad m = 20$$

Domains: tech, medical, business, financial, HR, legal, science, + multilingual (IT/FR/DE).

Personal preference domain is excluded — geometrically anti-aligned with SVD1.

Difference Matrix

Encode all texts, compute normalized difference vectors:

$$\mathbf{d}_i = \frac{\text{encode}(\text{new}_i) - \text{encode}(\text{old}_i)}{\|\text{encode}(\text{new}_i) - \text{encode}(\text{old}_i)\|_2}$$

Stack into matrix $\mathbf{D} \in \mathbb{R}^{m \times d}$.

SVD1 Direction

$$\mathbf{D} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$$

$$\mathbf{v}_{\text{svd1}} = \mathbf{V}_{1,:} \quad \text{(first right singular vector)}$$

Sign convention: orient $\mathbf{v}_{\text{svd1}}$ so the majority of seed pairs have positive alignment:

$$\mathbf{v}_{\text{svd1}} \leftarrow -\mathbf{v}_{\text{svd1}} \quad \text{if} \quad \frac{1}{m}\sum_i \langle \mathbf{d}_i, \mathbf{v}_{\text{svd1}} \rangle < 0$$

Direction Score

For a candidate (new, old) pair:

$$\text{dir}(\mathbf{e}_{\text{new}}, \mathbf{e}_{\text{old}}) = \left\langle \frac{\mathbf{e}_{\text{new}} - \mathbf{e}_{\text{old}}}{\|\mathbf{e}_{\text{new}} - \mathbf{e}_{\text{old}}\|_2}, \mathbf{v}_{\text{svd1}} \right\rangle$$

Returns `0.0` if `‖diff‖ < 10^{-8}` (paraphrase / near-identical).

Supersession Decision Logic

Given cosine similarity and direction score:

Condition	Action
Same scope, $\cos \geq 0.85$ + $\text{dir} \geq 0.10$	Supersede old schema
Same scope, $\cos \geq 0.85$ + $\text{dir} \in (-0.05, 0.05)$	Flag `needs_review`
Same scope, $\cos \geq 0.85$ + $\text{dir} < 0$	Reinforce old (contradiction)
Same scope, $\cos \in [0.70, 0.85)$ + $\text{dir} \geq 0.10$	Supersede (extended gate)
Cross-scope, $\cos \geq 0.78$	Reinforce + cross-scope evidence
$\cos \geq 0.35$	Mark topically related

Empirical Performance

- SVD1 axis separation (supersession vs. addition): **+0.35**
- Mean centroid separation: +0.09
- Cosine separation: +0.32
- SVD1 covers: tech, medical, business, financial, HR, legal, science, multilingual
- Personal preference: anti-aligned (–0.17) — excluded from seed set

Configuration (Global Constants)

Constant	Value	Description
`SAME_SCOPE_COS_THRESHOLD`	`0.85`	Cosine floor for same-scope action
`EXTENDED_SAME_SCOPE_COS_THRESHOLD`	`0.70`	Lower bound for extended same-scope gate
`CROSS_SCOPE_COS_THRESHOLD`	`0.78`	Cosine floor for cross-scope linking
`DIRECTION_THRESHOLD`	`0.10`	Minimum direction score for supersession
`DIR_REVIEW_BAND`	`(-0.05, 0.05)`	Direction range triggering needs_review
`TOPICAL_THRESHOLD`	`0.35`	Minimum similarity for topical relationship

Key Invariants

1. The SVD1 axis is computed once lazily and cached — `invalidate()` re-triggers computation (e.g., after encoder change).
2. Cross-scope schemas are **never** superseded — only reinforced with cross-scope evidence.
3. Direction score alone is never sufficient — must be combined with cosine ≥ threshold.
4. The extended gate (0.70–0.85) only triggers supersession when direction score is strongly positive — no reinforce or needs_review in this band.
5. SVD1 dimension normalization ensures all seed pairs contribute equally regardless of raw diff magnitude.